

Title: EFG-HF-PFS-gMG-2342 and CIDP-2401 Human Factor Studies	
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Purpose: This data on file packet is the reference for the following information: Data in this packet are from the two human factor studies: - EFG-HF-PFS-gMG-2342: Efgartigimod PH20 SC Prefilled Syringe for Generalized Myasthenia Gravis: Human Factors Validation Study - EFG-HF-PFS-CIDP-2401: Efgartigimod PH20 SC Prefilled Syringe for Chronic Inflammatory Demyelinating Polyneuropathy (CIDP): Human Factors Validation Study Data include study design information, participant disposition, demographic data, and results. These data supported the development of PFS for VYVGART Hytrulo and can help address questions around differences between the PFS and vial/winged infusion set, as well as any new safety concerns. V2.0: Annual renewal of DOF packet for continued use of supportive data from the PFS Human Factor Studies (gMG and CIDP) in medical and commercial materials. There are no changes to the data contained within DOF packet. Renewing DOF packet as the Hargraves et al. AAN 2025 Prefilled Syringe Poster provides only high-level information on the Human Factors Validation Studies in gMG and CIDP and does not include necessary context for the trial design and results..
Use(s): Please describe the manner in which the following data will be utilized (e.g., multiple assets, multiple agencies, etc.): These data may be used in the following example medical materials, including but not limited to: - MSL field materials - Medical Information documents - Frequently asked questions (FAQs) - Field readiness guide - Advisory boards - Symposia - HEOR materials These data may be used in the following example commercial materials, including but not limited to: - Pre-approval information exchange (PIE) and payer value proposition (PVP) decks - Commercial interactive visual aid (CIVA) - Frequently asked questions (FAQs) - HCP marketing materials - Patient marketing materials - Speaker decks - Printed leave behinds
Content: Data shown below are from the two human factor studies; data from the human factor study in patients with gMG and caregivers of patients with gMG or CIDP are presented first, followed by the study in patients with CIDP. Note: reference published materials (ex. the Borgions et al MGFA International 2025 poster titled “Investigating the Pharmacokinetics, Injection Speed, and Usability of Subcutaneous Efgartigimod PH20 Administration Using a Prefilled Syringe”) for a general overview of the study design. If applicable, reference study protocols for additional study design information.

EFG-HF-PFS-gMG-2342: Efgartigimod PH20 SC Prefilled Syringe for Generalized Myasthenia Gravis: Human Factors Validation Study**Study Design**

The main objectives of this study were to validate:

- Patients with gMG and caregivers of patients with gMG or CIDP could successfully use the efgartigimod PH20 SC prefilled syringe and associated instructional materials
- The efgartigimod PH20 SC prefilled syringe and associated instructional materials could be safely and effectively used, and that performance of critical tasks would not result in any use errors or harm to the patient, including compromised medical care
- Use-related hazards for the final/finished product have been mitigated and that any residual risks are acceptable (the benefit of product approval outweighs the residual risk)

The drug in this prefilled syringe is of high concentration (200 mg/mL) and an increased viscosity (11 cPs) compared to water for injection, and higher drug volume (5 mL), resulting in an average of 30 seconds injection time.

Inclusion/Exclusion Criteria

gMG participants were included in this study if they were:

- At least 18 years of age
- Diagnosed with gMG
- Naïve or experienced in delivering injections. Injection-naïve participants had no previous experience performing injections on self, others, animals, or other injection surrogates (e.g. manikin or skin pad)

Lay Caregiver participants were included in this study if they were:

- At least 18 years of age
- A caregiver to a family member with gMG or CIDP
- Naïve or experienced in giving injections. Injection-naïve participants had no previous experience performing injections on self, others, animals, or other injection surrogates (e.g. manikin or skin pad)

All participants were excluded from this study if they:

- Had participated in a prior study, within the last 6 months, using this proposed or a similar product; or had participated in previous evaluations (preliminary or formative) for this proposed product
- Worked for a pharmaceutical or medical device company, the sponsor (argenx), or the test company (Interface Analysis Associates)
- Had an occupation related to human factors, usability, industrial design, marketing, or pharmaceutical product development
- Did not read and speak English fluently

Summary of Potential Use Errors

- Drawing from a vial
 - Potential Use Error: Efgartigimod for SC administration is presented in a vial and the liquid must be withdrawn into a syringe for injection.
 - Product Mitigation: Syringe comes prefilled, reducing the numbers of use steps significantly and thereby also the chances for use errors.
- Needle attachment
 - Potential Use Errors: Attaching the needle at an angle and not fully securing the needle onto the syringe; both use errors would cause loss in drug when attempting to dose resulting in an underdose.
 - IFU Mitigation: Dedicated statements and illustrations instructing the user to twist on the needle until resistance is felt, indicating the direction that the user should turn to attach the needle, and how the syringe should look once the needle is fully attached.
- High viscosity and large medicine volume increase the injection time

- Potential Use Error: The drug in this prefilled syringe is of high concentration (200 mg/mL) and an increased viscosity (11 cPs) compared to water for injection, and higher drug volume (5 mL), resulting in an average of 30 seconds injection time. As a result, the user may feel some resistance as they press down on the plunger when injecting the medicine, and if the user attempts to push the plunger down quickly, it will make the plunger harder to press.
- Product Mitigation: Large thumb pad and wide finger flange to support users during the injection; preferentially a thin wall needle that allows viscous drug to pass more smoothly
- IFU Mitigation: Dedicated statements and illustrations with call-outs instructing the user to allow the prefilled syringe to warm to room temperature by letting it sit for at least 30 minutes before use, which lowers the viscosity, and call-outs emphasizing to press down on the plunger slowly, targeting a 20-30 second injection time
- Needle pricks post-injection
 - Potential Use Error: Needle pricks following injection.
 - Product Mitigation: Use of safety needle is recommended (containing a needle shield), reducing the risk of a needle stick. The needle can be attached to any size Luer Lock syringe and used for medication delivery by simply pulling backwards on the protective cover and pushing it back to safely lock the needle after use. A wide and textured finger pad on the cover allows for shielding the needle with a finger of the same hand holding the syringe. Audible and tactile confirmations enhance shielding process control.
 - IFU Mitigation: Dedicated statements and illustrations supporting users in proper use of the needle shield.

Study Procedure

Introduction: This study was conducted in a simulated-use environment which mimicked a home setting. Participants had access to instructions for use and materials supplied with the PFS but did not receive training. They were asked to store the PFS.

Unaided Injection: The following day, participants performed one unaided simulated injection into an injection pad. Participants placed the pad over their clothes; caregiver participants placed the pad on a manikin representing their patient. The performance measures for the simulated injection were as follows. All measures were considered successful if they were achieved, unless otherwise noted.

Storage tasks

- General storage
- Does not keep in original carton (successful if the participant keeps in the original carton)

Preparing to inject

- Remove carton from refrigerator
- Remove syringe from tray
- Check expiration date
- Check for damage
- Check appearance of the medicine
- Warm to room temperature
- Gather supplies
- Wash hands
- Remove needle from packaging
- Snap off PFS cap
- Attach the needle
- Choose injection site
- Clean injection site

Injecting

- Pull back needle shield
- Remove needle cap
- Throw away needle cap

- Pinch injection site
- Insert needle
- Release pinched skin
- Inject dose
- Remove needle from skin

Disposal

- Cover needle
- Throw away used syringe

Additional measures

- Safe use
- Wet injection (successful if the participant does not have a wet injection)
- Close calls (successful if there are no close calls)
- Observed instances of difficulty (successful if there are no instances of difficulty)
- Calls for help (successful if there is no call for help)
- Treat injection site, if needed

Knowledge and Performance Tasks: Participants also performed a knowledge task, where they were asked questions about information that was not easily observed during the procedure, including information about storage, what to check on or about the PFS, warming to room temperature, injection sites, and treating the injection site. Participants also completed a performance task where they were also asked to identify the expiration date on the syringe label and whether they would be able to dose successfully with the PFS (subjective feedback). Please see the knowledge tasks below with performance measures:

KNOWLEDGE TASK: In addition to what comes in the carton, what supplies, if any, would you need for your injection?

- Gather supplies – Does not identify to gather safety needle
- Gather supplies – Does not identify to gather alcohol swab
- Gather supplies – Does not identify to gather sterile gauze and/or bandage
- Gather supplies – Does not identify to gather sharps disposal container

KNOWLEDGE TASK: How many doses can you give with a single PFS?

- Dose – Does not identify PFS is single use
- Dose – Does not identify to not reuse syringe

KNOWLEDGE TASK: Where should you store this product?

- Storage – Does not identify to store in refrigerator between 36°F to 46°F (2°C to 8°C)

KNOWLEDGE TASK: What can you tell me about storage of the syringe related to the carton?

- Storage – Does not identify to keep in original carton

KNOWLEDGE TASK: What storage considerations should you take if children are in the household?

- Storage – Does not identify to keep out of reach of children

KNOWLEDGE TASK: Are there any specific ways that you should not store this product? If yes, what are they?

- Storage – Does not identify to not freeze PFS
- Storage – Does not identify to not store in direct sunlight

KNOWLEDGE TASK: What should you do if the PFS has been frozen?

- Storage – Does not identify to not use PFS that has been frozen

KNOWLEDGE TASK: What should you do if the PFS has been left in direct sunlight?

- Storage – Does not identify to not use PFS left in direct sunlight

KNOWLEDGE TASK: What, if anything, should you check on or about the syringe before using it?

- Check expiration date – Does not identify to check expiration date on the syringe label
- Check for damage – Does not identify to check the PFS for damage
- Check appearance of the medicine – Does not identify to check the appearance of the medicine

KNOWLEDGE TASK: What, if anything, should you do if the syringe is expired?

- Check expiration date – Does not identify to not use expired medication

KNOWLEDGE TASK: What, if anything, should you do if the syringe shows signs of damage?

- Check for damage – Does not identify to not use PFS if it appears damaged

KNOWLEDGE TASK: What should the liquid medicine in the PFS look like?

- Check appearance of the medicine – Does not identify the liquid should be clear to light yellow
- Check appearance of the medicine – Does not identify a little cloudiness is normal

KNOWLEDGE TASK: What, if anything, should you do if the medicine does not look the way it should?

- Check appearance of the medicine – Does not identify to not use PFS if the medicine is discolored or contains particles

KNOWLEDGE TASK: What, if anything, should you do with the syringe after you've taken it out of the refrigerator and checked that it is OK to use?

- Warm to room temperature – Does not identify to allow PFS to warm to room temperature (sit for at least 30 minutes)

KNOWLEDGE TASK: What can you tell me about warming the PFS any other way than left on the surface?

- Warm to room temperature – Does not identify to not warm the PFS in any other way

KNOWLEDGE TASK: What, if anything, should you do if the PFS has been at room temperature for longer than 4 hours?

- Warm to room temperature – Does not identify to not leave PFS at room temperature for longer than 4 hours

KNOWLEDGE TASK: When choosing an injection site, how does the last injection site play a role in that decision?

- Choose injection site – Does not identify to rotate injection site for each injection

KNOWLEDGE TASK: What areas of the skin should you avoid, or not inject into?

- Choose injection site – Does not identify to not inject into skin that is irritated, red, bruised, infected, or scarred
- Choose injection site – Does not identify to not inject into a vein

KNOWLEDGE TASK: What, if anything, should you do if you see blood at the injection site after you remove the needle from the skin?

- Treat Injection Site – Does not identify to treat the injection site with gauze pad or adhesive bandage, if needed

Performance Task

- Check Expiration Date – Does not identify expiration date on syringe label

Participant Disposition

15 participants with gMG and 15 lay caregivers to family members with gMG or CIDP enrolled in EFG-HF-PFS-gMG-2342. All 30 participants completed the study with no early discontinuations.

Participant Demographics^a		
Demographics	Participants with gMG (n=15)	Lay caregivers (n=15)
Injection experience, n (%)		
Experienced	9 (60)	7 (47)
Naïve	6 (40)	8 (53)
Sex, n (%)		
Male	4 (27)	5 (33)
Female	11 (73)	10 (67)
Age, mean (range)	57.1 (31-77)	52.1 (31-65)
Ethnicity, n (%)		
African American	3 (20)	4 (27)
Asian	2 (13)	1 (7)
Caucasian	9 (60)	8 (53)
Hispanic	1 (7)	2 (13)
Education, n (%)		
High school diploma	3 (20)	6 (40)
Associate degree	2 (13)	0 (0)
Bachelor's degree	5 (33)	5 (33)
Master's degree	2 (13)	2 (13)
Doctorate ^b	3 (20)	2 (13)
Occupation, n (%)		
Employed ^c	7 (47)	13 (87)
Unemployed	8 (53)	2 (13)
Dexterity/hand impairment, n (%)		
Mild	4 (27)	0
Moderate	2 (13)	0
No impairment	9 (60)	15 (100)
Visual impairment, n (%)		
Yes	11 (73)	1 (7)
No	4 (27)	14 (93)
Hearing impairment, n (%)		
Yes	2 (13)	0
No	13 (87)	15 (100)
Walking impairment, n (%)		
Yes (uses assistive walking aids)	2 (13)	0
No	13 (87)	15 (100)
Cognitive impairment, n (%)		
Yes	6 (40) ^d	0
No	9 (60)	15 (100)
English fluency, n (%)		
Yes	15 (100)	15 (100)
No	0	0

^aAll gMG patients (15/15, 100%) and lay caregivers (15/15, 100%) reported no prior participation in any study involving the product; and no studies involving a similar product within the last six months. ^bIncluded one juris doctorate in the participants with gMG and one PhD in the lay caregivers. ^cOf employed patients and caregivers, none reported working for a pharmaceutical or medical device company, argenx, or the test company (Interface Analysis Associates). ^dMild to moderate brain fog, mild ADD.

Results

Unaided Dose Findings-Performance Measures

Overall, 100% of participants (30/30) were successful in delivering the full dose. During the unaided injection, two use errors and one close call occurred that did not affect dose delivery. The 2 use errors were related to storage, and the close call was related to preparing to inject – see descriptions below.

Storage

All participants (30/30, 100%) stored the syringes in the refrigerator, and twenty-eight participants (28/30, 93%) kept the syringes in the original carton when doing so.

There were 2 use errors related to storage:

- When storing the product, P18 (Lay Caregiver, Injection Experienced) took the syringe trays out of the carton and stored them in the refrigerator, then stated that he would throw the box away. When asked to identify how the syringes should be stored in regard to their carton during the knowledge task questions, P18 correctly identified to keep the prefilled syringes in the original carton and noted that he had not done so. P18 also successfully identified to store the product in the refrigerator, and not store the syringes in direct sunlight. When debriefed, P18 stated he had purposefully removed the syringes from the carton to save space (in the refrigerator).
- When storing the product, P29 (Lay Caregiver, Injection Experienced) placed only the syringe trays in the refrigerator and left the carton on the table. When asked to identify how the syringes should be stored in regard to their carton during the knowledge task questions, P29 correctly identified to keep the prefilled syringes in the original carton. P29 also successfully identified to store the product in the refrigerator, and not store the syringes in direct sunlight. When debriefed, P29 stated that she'd thought the trays were the "carton."
- Both participants stored the syringes in the refrigerator and in the sealed trays, and correctly identified information related to storage temperature and protecting the product from light. Given that there would be no exposure to light, there is no risk associated with removing the trays from the carton for storage in the refrigerator. As such, it was concluded that no further mitigation is necessary.

Preparing to Inject

All participants (30/30, 100%) were successful in removing the carton from refrigerator, removing a syringe from its tray, checking the expiration date, checking for damage, checking the appearance of the medicine, warming the syringe to room temperature for a minimum of 30 minutes, gathering supplies, washing hands, removing the needle from its packaging, snapping off the PFS cap, attaching the needle, choosing the injection site, and cleaning the injection site.

There was 1 close call related to removing the needle from the packaging:

- P24 (Lay Caregiver, Injection Naïve) correctly and successfully removed the needle from its packaging as per Step 5.1 of the IFU (*Carefully open the needle package and remove the needle. Throw away the packaging into household trash*). P24 then went on to perform Step 5.2 of the IFU (*Bend the PFS cap to one side to snap it off and remove it from the PFS. Throw away the syringe cap*), removing the cap from the syringe. P24 then inexplicably returned to Step 5.1 and interpreted the step as instructing her to remove the needle cap. After removing the needle cap, she realized her mistake and disposed of that needle, obtained a new needle, attached it to the syringe (Step 5.3 of the IFU [*Hold the PFS in one hand, and attach the needle to the PFS by twisting it on (clockwise/to the right) until you feel resistance. The needle is now attached to the PFS*]) and carried on with the injection process. When debriefed, P24 stated that the instruction "remove the needle" in Step 5.1 ("Carefully open the needle packaging and remove the needle") confused her, because after removing the needle from the package and looking back at that step she then thought "removing the needle" meant removing the plastic pieces on the needle (i.e., shield, cap) and exposing the needle. P24 looked at the figure showing the needle parts which helped clarify her confusion. P24 suggested adding a statement to Step 5.1 stating "Do not remove the needle cap yet". She further stated that her confusion was due to first time use and having no prior injection experience, and stated she would get used to the procedure over time.
- HF Analysis: Unanticipated Close call. This is attributed to the user's unfamiliarity with injectable medications and the component parts. Note also that removing the needle cap occurs immediately after the next step of choosing and cleaning the injection site. The participant was able to recover and successfully deliver the unaided injection with no further issues.

Injecting

All participants (30/30, 100%) were successful in pulling back the needle shield, removing the needle cap, throwing away the needle cap, pinching the injection site, inserting the needle, releasing the pinched skin, injecting the full dose, and removing the needle from the skin. All participants successfully delivered the full dose, and no occurrences of a wet injection were observed.

Disposal

All participants (30/30, 100%) were successful in covering the needle with the needle shield and throwing away the used syringe.

Knowledge Tasks and Expiration Date Check Findings-Performance Measures

Participants were asked to locate and identify critical information in the IFU. A total of 19 knowledge task questions were asked during the study. Across these knowledge tasks, there were 28 different measures or pieces of information that participants had to know, understand, or identify. We observed an overall success rate of 100% (840/840); all 30 participants achieved all 28 measures. This high success rate demonstrates that the IFU adequately supports users in locating, understanding, and identifying critical information within the instructions and applying it to real-world use.

Participants were also asked to identify the expiration date on the syringe label. All participants (30/30, 100%) successfully identified the expiration date on the syringe label.

Subjective Feedback

When asked if they feel they would be able to successfully give injections with the product on a weekly basis, if needed, all participants responded that they would be able to successfully use the PFS.

Q: Do you feel that you would be able to successfully give injections with this product on a weekly basis if you needed to?

Adult Patient Participant Additional Comments (Stated Yes):

- P1 - Yes. The instructions are really good, they're clear and the visuals are really good.
- P8 - Yes, if I needed to. Once you get accustomed to it, it would be easier. I think the instructions are pretty easy to follow.
- P10 - Yes. I could do it though. It is just a long and slow injection, I might be concerned about moving the needle around. The instructions were straightforward.
- P11 - Yes. The instructions are very detailed and easy to follow.
- P15 - Yes. And the more you do it, it gets easier over time.

Lay Caregiver Participant Additional Comments (Stated Yes):

- P24 - Yes, especially once I'd gotten more comfortable. This was my first time using anything with an injection.
- P27 - Yes. It's a little intimidating but over time it would be less so.

Differences Between Use Group Findings

No substantial differences in performance were observed between patients or caregivers. Additionally, no substantial differences were observed regardless of prior injection experience.

Safety

Safety data were not measured in this validation study.

EFG-HF-PFS-CIDP-2401: Efgartigimod PH20 SC Prefilled Syringe for Chronic Inflammatory Demyelinating Polyneuropathy (CIDP): Human Factors Validation Study**Study Design**

The main objectives of this study were to validate that:

- Patients with CIDP could successfully use the efgartigimod PH20 SC prefilled syringe and associated instructional materials

- The efgartigimod PH20 SC prefilled syringe and associated instructional materials could be safely and effectively used, and that performance of critical tasks would not result in any use errors or harm to the patient, including compromised medical care
- Use-related hazards for the final/finished product have been mitigated and that any residual risks are acceptable (the benefit of product approval outweighs the residual risk)

The drug in this prefilled syringe is of high concentration (200 mg/mL) and an increased viscosity (11 cPs) compared to water for injection, and higher drug volume (5 mL), resulting in an average of 30 seconds injection time.

Inclusion/Exclusion Criteria

CIDP participants were included in this study if they were:

- At least 18 years of age
- Diagnosed with CIDP
- Naïve or experienced in delivering injections. Injection-naïve participants had no previous experience performing injections on self, others, animals, or other injection surrogates (e.g. manikin or skin pad)

All participants were excluded from this study if they:

- Had participated in a prior study, within the last 6 months, using this proposed or a similar product; or had participated in previous evaluations (preliminary or formative) for this proposed product
- Worked for a pharmaceutical or medical device company, the sponsor (argenx), or the test company (Interface Analysis Associates)
- Had an occupation related to human factors, usability, industrial design, marketing, or pharmaceutical product development
- Did not read and speak English fluently

Summary of Potential Use Errors

- Drawing from a vial
 - Potential Use Error: Efgartigimod for SC administration is presented in a vial and the liquid must be withdrawn into a syringe for injection.
 - Product Mitigation: Syringe comes prefilled, reducing the numbers of use steps significantly and thereby also the chances for use errors.
- Needle attachment
 - Potential Use Errors: Attaching the needle at an angle and not fully securing the needle onto the syringe; both use errors would cause loss in drug when attempting to dose resulting in an underdose.
 - IFU Mitigation: Dedicated statements and illustrations instructing the user to twist on the needle until resistance is felt, indicating the direction that the user should turn to attach the needle, and how the syringe should look once the needle is fully attached.
- High viscosity and large medicine volume increase the injection time
 - Potential Use Error: The drug in this prefilled syringe is of high concentration (200 mg/mL) and an increased viscosity (11 cPs) compared to water for injection, and higher drug volume (5 mL), resulting in an average of 30 seconds injection time. As a result, the user may feel some resistance as they press down on the plunger when injecting the medicine, and if the user attempts to push the plunger down quickly, it will make the plunger harder to press.
 - Product Mitigation: Large thumb pad and wide finger flange to support users during the injection; preferentially a thin wall needle that allows viscous drug to pass more smoothly
 - IFU Mitigation: Dedicated statements and illustrations with call-outs instructing the user to allow the prefilled syringe to warm to room temperature by letting it sit for at least 30 minutes before use, which lowers the viscosity, and call-outs emphasizing to press down on the plunger slowly, targeting a 20-30 second injection time
- Needle pricks post-injection
 - Potential Use Error: Needle pricks following injection.

- Product Mitigation: Use of safety needle is recommended (containing a needle shield), reducing the risk of a needle stick. The needle can be attached to any size Luer Lock syringe and used for medication delivery by simply pulling backwards on the protective cover and pushing it back to safely lock the needle after use. A wide and textured finger pad on the cover allows for shielding the needle with a finger of the same hand holding the syringe. Audible and tactile confirmations enhance shielding process control.
- IFU Mitigation: Dedicated statements and illustrations supporting users in proper use of the needle shield.

Study Procedure

Introduction: This study was conducted in a simulated-use environment which mimicked a home setting. Participants had access to instructions for use and materials supplied with the PFS, but did not receive training. They were asked to store the PFS.

Unaided Injection: The following day, participants performed one unaided simulated injection into an injection pad. Participants placed the pad over their clothes. The performance measures for the simulated injection were as follows. All measures were considered successful if they were achieved, unless otherwise noted.

Storage tasks

- General storage
- Does not keep in original carton (successful if the participant keeps in the original carton)

Preparing to inject

- Remove carton from refrigerator
- Remove syringe from tray
- Check expiration date
- Check for damage
- Check appearance of the medicine
- Warm to room temperature
- Gather supplies
- Wash hands
- Remove needle from packaging
- Snap off PFS cap
- Attach the needle
- Choose injection site
- Clean injection site

Injecting

- Pull back needle shield
- Remove needle cap
- Throw away needle cap
- Pinch injection site
- Insert needle
- Release pinched skin
- Inject dose
- Remove needle from skin

Disposal

- Cover needle
- Throw away used syringe

Additional measures

- Safe use
- Wet injection (successful if the participant does not have a wet injection)
- Close calls (successful if there are no close calls)
- Observed instances of difficulty (successful if there are no instances of difficulty)
- Calls for help (successful if there is no call for help)

- Treat injection site, if needed

Knowledge and Performance Tasks: Participants also performed a knowledge task, where they were asked questions about information that was not easily observed during the procedure, including information about storage, what to check on or about the PFS, warming to room temperature, injection sites, and treating the injection site. Participants also completed a performance task where they were also asked to identify the expiration date on the syringe label and whether they would be able to dose successfully with the PFS (subjective feedback). Please see the knowledge tasks below with performance measures:

KNOWLEDGE TASK: In addition to what comes in the carton, what supplies, if any, would you need for your injection?

- Gather supplies – Does not identify to gather safety needle
- Gather supplies – Does not identify to gather alcohol swab
- Gather supplies – Does not identify to gather sterile gauze and/or bandage
- Gather supplies – Does not identify to gather sharps disposal container

KNOWLEDGE TASK: How many doses can you give with a single PFS?

- Dose – Does not identify PFS is single use
- Dose – Does not identify to not reuse syringe

KNOWLEDGE TASK: Where should you store this product?

- Storage – Does not identify to store in refrigerator between 36°F to 46°F (2°C to 8°C)

KNOWLEDGE TASK: What can you tell me about storage of the syringe related to the carton?

- Storage – Does not identify to keep in original carton

KNOWLEDGE TASK: What storage considerations should you take if children are in the household?

- Storage – Does not identify to keep out of reach of children

KNOWLEDGE TASK: Are there any specific ways that you should not store this product? If yes, what are they?

- Storage – Does not identify to not freeze PFS
- Storage – Does not identify to not store in direct sunlight

KNOWLEDGE TASK: What should you do if the PFS has been frozen?

- Storage – Does not identify to not use PFS that has been frozen

KNOWLEDGE TASK: What should you do if the PFS has been left in direct sunlight?

- Storage – Does not identify to not use PFS left in direct sunlight

KNOWLEDGE TASK: What, if anything, should you check on or about the syringe before using it?

- Check expiration date – Does not identify to check expiration date on the syringe label
- Check for damage – Does not identify to check the PFS for damage
- Check appearance of the medicine – Does not identify to check the appearance of the medicine

KNOWLEDGE TASK: What, if anything, should you do if the syringe is expired?

- Check expiration date – Does not identify to not use expired medication

KNOWLEDGE TASK: What, if anything, should you do if the syringe shows signs of damage?

- Check for damage – Does not identify to not use PFS if it appears damaged

KNOWLEDGE TASK: What should the liquid medicine in the PFS look like?

- Check appearance of the medicine – Does not identify the liquid should be clear to light yellow

- Check appearance of the medicine – Does not identify a little cloudiness is normal

KNOWLEDGE TASK: What, if anything, should you do if the medicine does not look the way it should?

- Check appearance of the medicine – Does not identify to not use PFS if the medicine is discolored or contains particles

KNOWLEDGE TASK: What, if anything, should you do with the syringe after you've taken it out of the refrigerator and checked that it is OK to use?

- Warm to room temperature – Does not identify to allow PFS to warm to room temperature (sit for at least 30 minutes)

KNOWLEDGE TASK: What can you tell me about warming the PFS any other way than left on the surface?

- Warm to room temperature – Does not identify to not warm the PFS in any other way

KNOWLEDGE TASK: What, if anything, should you do if the PFS has been at room temperature for longer than 4 hours?

- Warm to room temperature – Does not identify to not leave PFS at room temperature for longer than 4 hours

KNOWLEDGE TASK: When choosing an injection site, how does the last injection site play a role in that decision?

- Choose injection site – Does not identify to rotate injection site for each injection

KNOWLEDGE TASK: What areas of the skin should you avoid, or not inject into?

- Choose injection site – Does not identify to not inject into skin that is irritated, red, bruised, infected, or scarred
- Choose injection site – Does not identify to not inject into a vein

KNOWLEDGE TASK: What, if anything, should you do if you see blood at the injection site after you remove the needle from the skin?

- Treat Injection Site – Does not identify to treat the injection site with gauze pad or adhesive bandage, if needed

Performance Task

- Check Expiration Date – Does not identify expiration date on syringe label

Participant Disposition

15 participants with CIDP enrolled in EFG-HF-PFS-CIDP-2401. All 15 participants completed the study with no early discontinuations.

Participant Demographics^a

Demographics	Participants with CIDP (N=15)
Injection experience, n (%)	
Experienced	6 (40)
Naïve	9 (60)
Sex, n (%)	
Male	8 (53)
Female	7 (47)
Age, mean (range)	43.1 (19-70)
Ethnicity, n (%)	
African American	

Caucasian	5 (33)
Hispanic	9 (60)
	1 (7)
Education, n (%)	
High school diploma	6 (40)
Vocational certificate	1 (7)
Associate degree	1 (7)
Bachelor's degree	6 (40)
Master's degree	1 (7)
Occupation, n (%)	
Employed ^b	6 (40)
Unemployed	9 (60)
Dexterity/hand impairment, n (%)	
Mild	2 (13)
Moderate	3 (20)
No impairment	10 (67)
Visual impairment, n (%)	
Yes	4 (27)
No	11 (73)
Hearing impairment, n (%)	
Yes	1 (7)
No	14 (93)
Walking impairment, n (%)	
Yes (uses assistive walking aids)	5 (33)
No	10 (67)
Cognitive impairment, n (%)	
Yes	9 (60)
No	6 (40)
English fluency, n (%)	
Yes	15 (100)
No	0

^aAll CIDP patients (15/15, 100%) reported no prior participation in any study involving the product; and no studies involving a similar product within the last six months. ^bOf employed patients and caregivers, none reported working for a pharmaceutical or medical device company, argenx, or the test company (Interface Analysis Associates).

Results

Unaided Dose Findings-Performance Measures

All participants (15/15, 100%) successfully prepared and delivered the full dose. Additionally, participants had no difficulty handling the syringe, or pushing the plunger down to deliver the full dose. This indicates that the instruction to push down on the plunger slowly was effective in reducing the plunger movement forces and maintaining a comfortable injection.

Performance of critical tasks did not result in any patterns of use errors, close calls, or difficulties that would lead to patient harm (including compromised medical care).

Storage

All participants (15/15, 100%) stored the syringes in the refrigerator, and all participants (15/15, 100%) kept the syringes in the original carton when doing so.

Preparing to Inject

All participants (15/15, 100%) were successful in removing the carton from refrigerator, removing a syringe from its tray, checking the expiration date, checking for damage, checking the appearance of the medicine, warming the syringe to room temperature for a minimum of 30 minutes, gathering supplies, washing hands, removing the needle from its packaging, snapping off the prefilled syringe cap, attaching the needle, choosing the injection site, and cleaning the injection site.

Injecting

All participants (15/15, 100%) were successful in pulling back the needle shield, removing the needle cap, throwing away the needle cap, pinching the injection site, inserting the needle, releasing the pinched skin, injecting the full dose, and removing the needle from the skin.

All participants successfully delivered the full dose, and no occurrences of a wet injection were observed.

Disposal

All participants (15/15, 100%) were successful in covering the needle with the needle shield and throwing away the used syringe.

Knowledge Tasks and Expiration Date Check Findings-Performance Measures

Participants were asked to locate and identify critical passages in the IFU. A total of 19 knowledge task questions were asked during the study. Across these knowledge tasks, there were 28 different measures or pieces of information that participants had to know, understand, or identify. We observed an overall success rate of 100% (420/420); all 15 participants achieved all 28 measures. This high success rate demonstrates that the IFU adequately supports users in locating, understanding, and identifying critical information within the instructions and applying it to real-world use.

Additionally, participants were asked to identify the expiration date on the syringe label. All participants (15/15, 100%) successfully identified the expiration date on the PFS label.

Subjective Feedback

Participant feedback was positive, with all participants (15/15, 100%) stating that they could safely and successfully prepare and dose with this product. Participants also found the IFU to be helpful and supportive.

Q: Do you feel that you would be able to successfully give injections with this product on a weekly basis if you needed to?

Adult Patient Participant Comments:

- P1 - Yes.
- P2 - Yes. This is a lot better than getting an IV infusion for 6 hours.
- P3 - Yes.
- P4 - Yes.
- P5 - Yes, this is much easier than what I use now.
- P6 - Yes.
- P7 - Yes. I think they do a good job of explaining everything you need to do in the instructions, it's pretty straightforward. I like the colors, the pictures are very intuitive, I think it's a very helpful instruction guide.
- P8 - Yes, but it was a little hard to push in so long with my hands, but it's doable.
- P9 - Yes, I love the way the instructions are set up.
- P10 - Yes. I really like all of the diagrams and figures in the instructions, that's super helpful. It helps to break it down and follow step by step so that's really awesome.
- P11 - Yes, with the instructions, absolutely. It's pretty straightforward.
- P12 - Yes, absolutely. It's so much easier than what I have to do now. It's much easier than an infusion. The slow and long injection isn't an obstacle for me at all. If this works, this is a game changer.
- P13 - Yes, it would just take a little getting used to.
- P14 - Yes. Everything is clear. And you would get the hang of it after a while. I like the big clear illustrations; it makes it easy to see what to do.
- P15 - Yes. It's fairly straightforward.

Safety

Safety data were not measured in this validation study.

References:

▪ **EFG-HF-PFS-gMG-2342 (Human Factors Study - gMG)**

- EFG-HF-PFS-gMG-2342 Clinical Study Report March 7, 2024, including the following sections:
 - 6.1: Study Objectives
 - 6.5.1.2: Key Demographic Sub-Factors and Co-Morbidities
 - 4.7: Summary of Known Use Problems
 - 6.10: Study Procedure
 - 6.6.3: Unaided Tasks
 - 6.5.3: Summary of Demographic Data
 - 7.1: Unaided Dose Findings
 - 7.2: Knowledge Tasks and Expiration Date Check Findings
 - 7.3: Final Disposition Findings

▪ **EFG-HF-PFS-CIDP-2401 (Human Factors Study – CIDP)**

- EFG-HF-PFS-CIDP-2401 Clinical Study Report March 7, 2024, including the following sections:
 - 6.1: Study Objectives
 - 6.5.1.2: Key Demographic Sub-Factors and Co-Morbidities
 - 4.7: Summary of Known Use Problems
 - 6.10: Study Procedure
 - 6.6.3: Unaided Tasks
 - 6.5.3: Summary of Demographic Data
 - 7.1: Unaided Dose Findings
 - 7.2: Knowledge Tasks and Expiration Date Check Findings
 - 7.3: Final Disposition Findings

Talking Points:

Overall

Refer to published materials (ex. Borgions et al MGFA International 2025 poster) for high-level results:

- Human factor validation studies demonstrated that participants with gMG or CIDP and lay caregivers can safely and successfully prepare and administer efgartigimod PH20 SC PFS
- 100% of participants (N=30/30 in the gMG study; N=15/15 in the CIDP study) were successful in preparing and delivering the full dose in an average of 30 seconds. Participants and lay caregivers had no difficulty handling the syringe and successfully identified critical information on the instructions. Residual risks were as low as possible and were not tied to the design of the prefilled syringe or instructional materials

Additional points below from each study.

EFG-HF-PFS-gMG-2342

- This study demonstrated that both participants with gMG and lay caregivers can safely and successfully prepare and administer efgartigimod PH20 SC PFS and interact with the associated labeling, packaging, and instructional materials.
- 30/30 subjects provided subjective feedback that they feel they were able to perform the injection (no training provided) and injection time between 20-30 seconds is achievable
- The associated instructional materials are straightforward, understandable, and translatable into successful physical actions by the intended user populations
- Performance of critical tasks did not result in any patterns of use errors, close calls, or difficulties that would lead to patient harm (including compromised medical care).
- The use-related hazards have been adequately mitigated; there are no residual risks to be further mitigated. All risks are at acceptable levels. Residual risks are as low as possible and not tied to the design of the prefilled syringe or instructional materials.

The benefits of Efgartigimod PH20 SC PFS outweigh any residual risks and are likely to have a positive impact on the intended patient population (adult patients with gMG).

EFG-HF-PFS-CIDP-2401

- This study demonstrated that participants with CIDP can safely and successfully prepare and administer efgartigimod PH20 SC PFS and interact with the associated labeling, packaging, and instructional materials.
- 15/15 subjects provided subjective feedback that they feel they were able to perform the injection (no training provided) and injection time between 20-30 seconds is achievable
- The associated instructional materials are straightforward, understandable, and translatable into successful physical actions by the intended user populations
- Performance of critical tasks did not result in any patterns of use errors, close calls, or difficulties that would lead to patient harm (including compromised medical care)
- The use-related hazards have been adequately mitigated; there are no residual risks to be further mitigated. All risks are at acceptable levels. Residual risks are as low as possible and not tied to the design of the prefilled syringe or instructional materials.
- The benefits of Efgartigimod PH20 SC PFS outweigh any residual risks and are likely to have a positive impact on the intended patient population (adult patients with CIDP)
-

EFG-HF-PFS-gMG-2342

Checklist for additional requirements to appropriately describe analyses:

■ Type of analysis-Human factors study

☐ Analysis set

☐ Mean at baseline

☐ Confounders, if applicable

■ Limitations of analysis-Findings were based on performance, observed behaviors, subjective feedback and Human Factors analyses, and therefore were not traditionally statistically analyzed.

EFG-HF-PFS-CIDP-2401

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Approval

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